What Drives Asset Pricing Factor Returns? Evidence from Machine Learning

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Motivation

- Money in factor investment is big
  - US factor style ETF assets were at $482 billion by 2019 with 78% growth from new flows according to Andrew Ang (2019)
- Since factor returns vary over time, investors care about whether timing factors brings economic benefits
- Numerous asset managers from leading asset management companies are responding to questions on factor timing profits
  - Contrarian factor timing is deceptively difficult by Asness et al. (2017)
  - Timing adds value to factor strategies, tricky but possible by Bender et al. (2018)
  - Is value investing dead? by Asness (2020)
  - Value's death may be greatly exaggerated by Arnott et al. (2020)

Motivation

Understanding factor predictability is also important for academia:

- While theories sharply disagree about what drives non-market factors, predictability can help identify the driving force
- Characteristics-based factors are also returns, and returns are driven by movements of cash flow or discount rate. The return decomposition by Campbell and Shiller (1988) can be applied to those factors directly.
  - Unexpected Return (return news) = Cash Flow News (CF) - Discount Rate News (DR)
  - Model conditional expected returns as a one-period forward prediction, and model CF by adding return news and DR, hence **results are sensitive to return predictability**

Cases when Decomposition Results are Distorted

- **Underestimated predictive power:** if the prediction model does not have any predictive power, then conditional expected return is a constant, all return shocks are left to CF
- **Overestimated predictive power:** if predictions fit returns almost perfectly (usually in sample), CF and DR could be highly positive correlated, large and volatile compared to return news; no return shocks in perfect fit case

Challenges

- In practice, CF and DR are usually estimated by Vector Autoregressions (VAR)
- With two properties by nature, VAR is a model that takes risks of distorting decomposition in the above cases
  - Linearity
    - It is doubtful that the relation between equity premium and state variables is linear **(underestimated)**
    - Standard linear regression breaks down when number of predictors is large or highly correlated **(underestimated)**
  - In-sample estimate
    - In-sample mean squared error (MSE) is essentially an underestimated statistics of out-of-sample MSE **(overestimated)**

What I Do

- How to do better in the prediction stage?
  - Linearity and in-sample properties are difficult to change under VAR
- Add or change predictors in VAR
  - Potential misspecifications is still there
- Is there a combined predictor with all predictive power from other standard predictors?
  - No need to subjectively choose predictors
  - Supervised machine learning techniques help to aggregate predictors

What I Do

- I construct a new predictor of factor returns for VAR by using machine learning techniques
- Prediction stage: predict factor returns out-of-sample by machine learning
- Decomposition stage: use the prediction made in the prediction stage as the only predictor in VAR for factor return decomposition
  - Fair predictive power

Main Empirical Findings

- Factor returns are predicable
  - 5 factors from Fama and French (2015) plus momentum are examined
  - Predictability varies within different training and testing sample pairs
  - Linear models of expected return should not be abandoned
- Cash flow news drives all six factor returns, its correlation with discount rate news is highly positive
  - Cast doubt on theories that common discount rate shocks drive returns via time-varying risk aversions or time-varying investor sentiments
  - Cast doubt on theories and empirical evidence suggesting negative correlations between DR and CF
- Shocks driving macroeconomy, market, and non-market factors are distinct
  - Correlations between components of market factor shocks and non-market factor shocks are weak, and those components correlate weakly with macroeconomic shocks

Related Literature

- Studies following Campbell and Shiller's (1988) decomposition to analyze return movements
  - Campbell (1991) for aggregate market return
  - Vuolteenaho (2002) for firm-level stocks
  - Campbell and Vuolteenaho (2004a) for size and value anomalies
  - Chen and Zhao (2009) examine the sensitivity of this method
  - Lochstoer and Tetlock (2020) for anomaly returns

- Studies employing nontrivial models to predict returns
  - Campbell and Thompson (2008), Rapach, Strauss, and Zhou (2010) adjust standard linear forecasts
  - Kelly and Pruitt (2013), Rossi (2018), Coulombe et al. (2019), Gu, Kelly and Xiu (2020) use machine learning techniques

- Studies regarding to asset pricing factors predictability
  - Cohen, Polk and Vuolteenaho (2003), Daniel and Moskowitz (2016), Haddad, Kozak, and Santosh (2020)

Assumptions

- Unlike the machine learning paper by Gu et al. (2020), which aggregates individual stock predictions for portfolio return predictions, I directly predict factor returns by aggregate predictors for each long and short portfolios

- The basic assumption is that **the next period of factor returns can be predicted by some function forms with aggregate predictors for the corresponding long and short portfolios in the current period**

$$\begin{aligned}\mathbb{E}_t[R^{f_i}_{t+1}] &= \mathbb{E}_t[R^{L_i}_{t+1}] - \mathbb{E}_t[R^{S_i}_{t+1}] \\ &= g_t[Z^{L_i}_t] - g_t[Z^{S_i}_t]\end{aligned}$$

g_t denotes some function forms which are re-estimated every period, Z_t denotes portfolio-level predictors, R^{f_i} denotes factor returns

Prediction Models

- There are numerous models to predict returns, I typically choose 3 with the following capabilities to focus on tackling potential misspecifications (limited predictors, linearity, in-sample) of VAR
  - They work well when the number of predictors is large, generally via some variable selection or variance reduction channels
  - They are models for both linear and nonlinear relations between RHS and LHS
  - They are shown to predict returns out-of-sample
- Models I use:
  - Combination Forecasts
  - Shrinkage Linear Regressions (Ridge/Lasso)
  - Boosted Regression Trees

Prediction Models: Combination Forecast

- Combination forecast:

$$\hat{R}_{t+1} = \sum_{k=1} \omega_{k,t} \hat{R}_{k,t+1}$$

where $\hat{R}_{k,t+1} = \hat{\alpha}_{k,t} + \hat{\beta}_{k,t} Z_{k,t}$, $Z_{k,t}$ is a single predictor k at time t; here I assume ω to be equal weights

- Though its simplicity, it reflects a widely applied thought in machine learning which is variance reduction to individual linear forecasts
  - The counterparts in nonlinear models are tree-based models, which are essentially an average of decision tree forecasts

Prediction Models: Shrinkage Linear Regression

- Shrinkage Linear Regression (Ridge and Lasso):
  - The forms are like linear multivariate regressions $\hat{R}_{t+1} = \beta_{0,t} + \sum_{k=1} \beta_{k,t} Z_{k,t}$, however, it adds a penalty term to residual sum of squared (RSS) in the loss function
    - So it measures a linear relation with mitigation of overfitting
  - For example, loss functions for Ridge and Lasso are:

$$RSS + \theta \sum_{k=1} \beta_{k,t}^2 \qquad RSS + \theta \sum_{k=1} |\beta_{k,t}|$$

    respectively, where RSS is the loss function in OLS case: $||Y - X\beta||_2^2$, θ is a tuning parameter regarding to shrinking parameters to what extent; the tuning parameters are selected by 10-fold cross validation within the training sample
  - A key advantage of Lasso over Ridge is that Lasso explicitly select some variables by shrinkage others parameters to be 0

Prediction Models: BRT

- Boosted Regression Trees: BRT is an nonlinear and semi-parametric model derived from a single decision tree
  - A decision tree is to do recursive binary partitions according to a dependent variable which can be written as $f(x) = \sum_{j=1}^{J} c_j I\{x \in S_j\}$
  - However, decision trees are rarely used in practice because structures of single trees usually changes dramatically from sample to sample, so its predictive power is low
  - Booting is an algorithm to grow tree sequentially, then averaging forecasts from many tree models. Essentially, BRT is a refined decision tree model, one of the first paper providing the algorithms is Freund and Shapire (1995)
  - When boosting, a new tree is developed by fitting the residuals of a previous tree, the speed of fitting is controlled by a tuning parameter; a small tuning parameter means only fitting a small fraction of residuals at each time. For parameters like this, I follow Rossi (2018)'s specifications

Predictors

- The predictors I use are monthly characteristic-based stock return predictors and macroeconomic predictors for aggregate market
  - Characteristic-based predictors contain 27 variables related to trend, liquidity, and volatility examined by Gu, Kelly, and Xiu (2020) which are shown to be more important predictors relatively among over 100 candidate conditioning variables for individual stock returns
  - I also contain 7 characteristic predictors, which are return, ROE, BM ratio, profitability, Investment, Size, and 6 month momentum used in Lochstoer and Tetlock (2020) for in-sample anomalies predictions
  - 14 macroeconomic predictors suggested by academia for market returns are also included, which are collected by Welch and Goyal (2008)

Predictors

- In total, 45 predictors are chosen as inputs for my prediction models (some from the three different sources are the same)
- Characteristics predictors are for individual stocks, so I aggregate them by value-weighting to long and short portfolio levels : $Z_t = \sum w_i Z_{it}$ where Z_t is a predictor for a long or a short portfolio, Z_{it} are predictors for individual stocks among that portfolio
- Macroeconomic predictors are directly used from Goyal's website as predictors

Predictors: Characteristic-based

1 month mom	Size	Change in mom	Maxium return
Volatility	Tradingi volume	Sale/Price	Turnover
Investment	Earning Increase	R&D	6 month mom
36 month mom	Earning/Price	Share outstanding	Secured debt
Idiosyncratic volatility	Bid-ask spread	Illiquidity	Age
Convertible debt	Beta	Beta-square	Cash productivity
Zero-trading days	Dividend yield	SG&A expenses	ROE
Profitability	Return	Book/Market	

Predictors: Macroeconomic

Dividend/Price	Earning/Price	Book/Market	Net Equity Expansion
Treasury bill yield	Default spread	Stock variance	Dividend/earning
Dividend yield	Term spread	Inflation	Risk free rate
LT bond yield	LT bond return		

Evaluation: OOS R-squared

- $$R^2_{oos} = 1 - \frac{\sum_{t \in testing}(y_{t+1} - \hat{y}_{t+1})^2}{\sum_{t \in testing}(y_{t+1} - \bar{y}_{training})^2}$$

  where training sample average are updated every period.

- Small positive R-squared can map into large utility gains for a mean-variance investor, Campbell and Thompson (2008)

Evaluation: Pseudo-out-of-sample R-squared Test

- This test is motivated by Coulombe et al. (2019), and is consistent with Diebold and Mariano (1995) test
- Assume squared prediction error can be decomposed as:

$$e_{t,m}^2 = \alpha_m + e_{t,b}^2 + v_{t,m}$$

where $e_{t,b}^2$ is the squared error of a benchmark model

- Then transform both LHS and RHS, for example, $e_{t,m}^2$ is transformed as $R_{oos,t,m}^2 = 1 - \frac{e_{t,m}^2}{\frac{1}{T}\sum_{t=1}^{T}(y_{t+1}-\bar{y}_t)^2}$, so that the average of $R_{oos,t,m}^2$ is OOS R-squared, and the error decomposition is:

$$R_{oos,t,m}^2 - \dot{e}_{t,b}^2 = \dot{\alpha}_m + \dot{v}_{t,m}$$

$\dot{\alpha}_m$ is interpreted as marginal improvements in OOS R-squared, and it means an improvement over benchmark model if positive

Campbell and Shiller's Decomposition

- Combining the present value relation and return log-linearization yields a decomposition of return shocks, which I directly apply to each factor time series:

$$r_{t+1}^{f_i} - \mathbb{E}_t[r_{t+1}^{f_i}] = \underbrace{[\mathbb{E}_{t+1} - \mathbb{E}_t](\sum_{j=1}^{\infty} \kappa_1^j g_{t+j}^{f_i})}_{CF_{t+1}^{f_i}} - \underbrace{[\mathbb{E}_{t+1} - \mathbb{E}_t](\sum_{j=2}^{\infty} \kappa_1^j r_{t+j}^{f_i})}_{DR_{t+1}^{f_i}}$$

A Bivariate VAR Approach

- A standard way to estimate the above decomposition is by VAR with the form:

$$X_{t+1} = AX_t + \eta_{t+1}$$

where $X_t = [r_t; Z_t]$, Z contains return predictors, for example, Z contains p-d ratio and real risk free rate in Campbell (1991) for aggregate market returns

- In the bivariate VAR setting, we have

$$X_t = [r_t^{f_i}; E_t[r_{t+1}^{f_i}]]$$

where $E_t[r_{t+1}^{f_i}]$ can be seen as a combination of predictors

A Bivariate VAR Approach

- The estimates of discount rate news and cash flow news are:

$$DR_{t+1}^{f_i} = e_1' \kappa A^{f_i} (I - \kappa A^{f_i})^{-1} \eta_{t+1}^{f_i} = e_1' \lambda^{f_i} \eta_{t+1}^{f_i}$$

$$CF_{t+1}^{f_i} = e_1' [I + \kappa A^{f_i} (I - \kappa A^{f_i})^{-1}] \eta_{t+1}^{f_i} = e_1' (I + \lambda^{f_i}) \eta_{t+1}^{f_i}$$

- Corresponding return variance decomposition is:

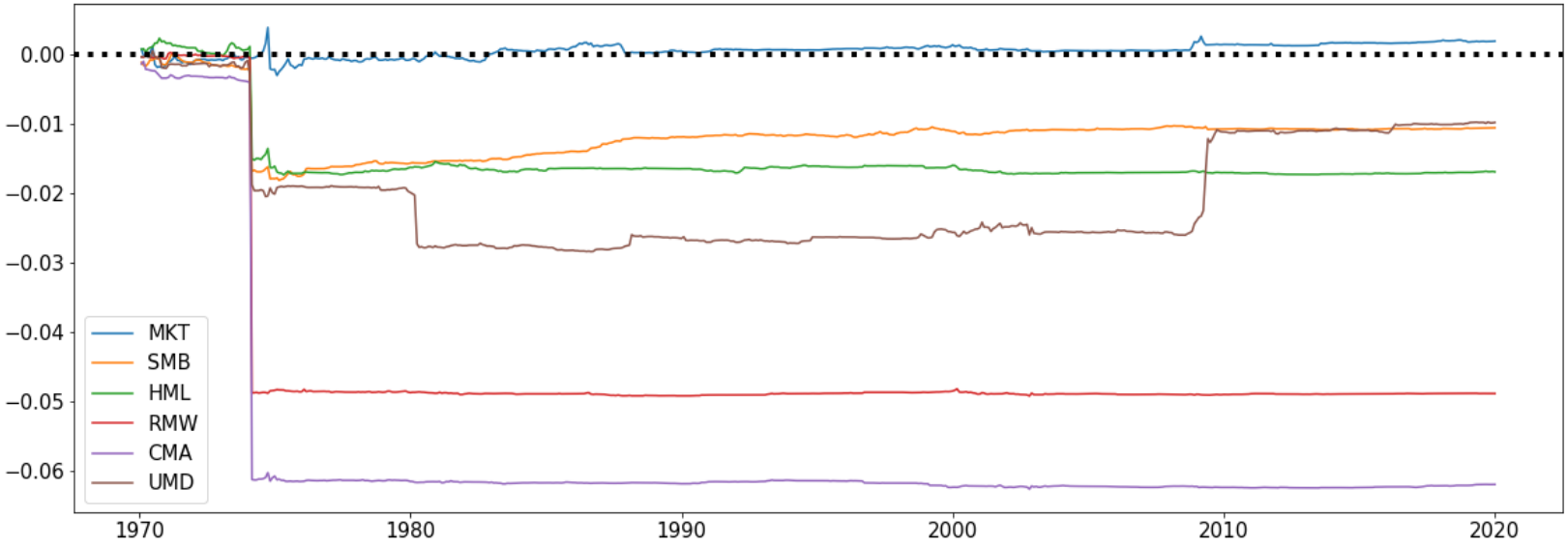
$$\sigma^2(DR^{f_i}) = e_1' \lambda^{f_i} \sigma^2(\eta^{f_i}) (e_1' \lambda^{f_i})'$$
$$\sigma^2(CF^{f_i}) = e_1' (I + \lambda^{f_i}) \sigma^2(\eta^{f_i}) (e_1' (I + \lambda^{f_i}))'$$
$$-2cov(DR^{f_i}, CF^{f_i}) = \sigma^2(\eta^{f_i}) - \sigma^2(DR^{f_i}) - \sigma^2(CF^{f_i})$$

Data

- Monthly factors returns are those in Fama and French (2015) plus momentum which are MKT, SMB, HML, RMW, CMA, and UMD.
  - Non-market factors constructions follow descriptions of factors formed on 2×3 portfolios from French's website.
- Data related to security price, volume, and firm-level characteristics are from CRSP and COMPUSTAT
- Macroeconomic predictors for returns are public available in Goyal's website
- Data related to monthly macroeconomic indicators are from FRED-MD data set, which is available from the website of Federal Reserve Bank of St. Louis
- Period from 01/1968 to 12/2019, training samples starts as 01/1968 - 01/1970, and expands every period

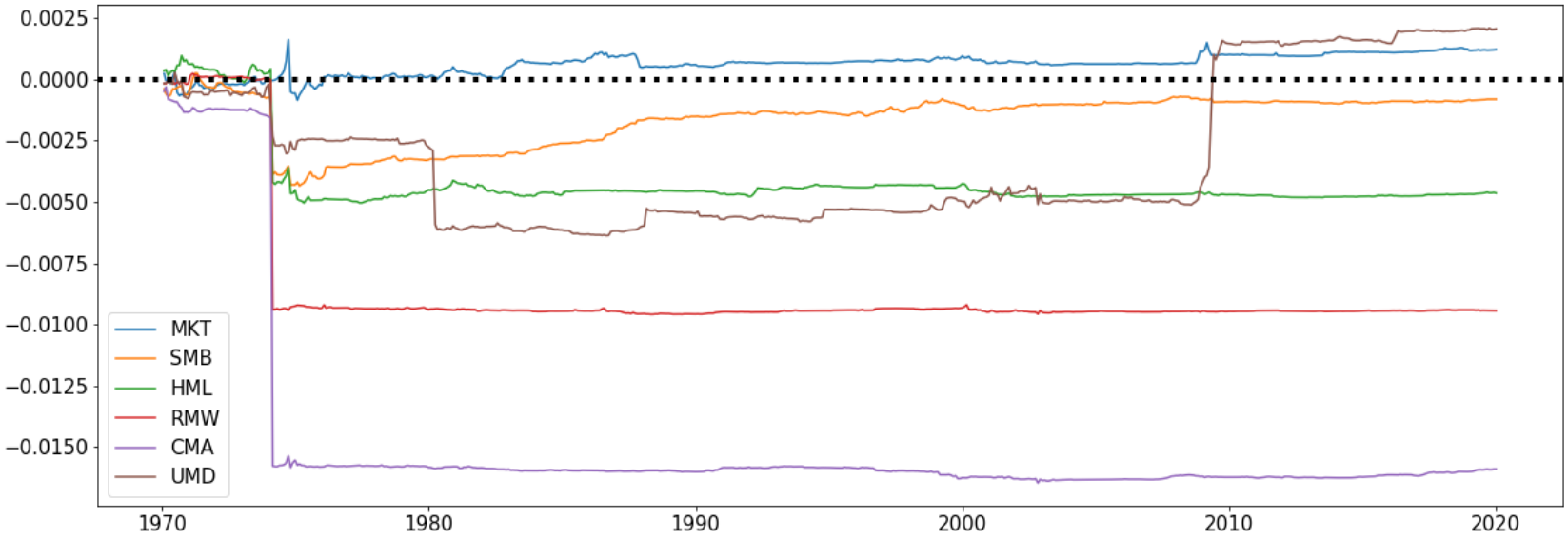
Predictability Visualization

- Difference of cumulative SSE between the Combination Forecast and the Naive Forecast
  - Cumulative SSE of the Naive Forecast - Cumulative SSE of the Combination Forecast
  - Above zero line, the compared model outperforms naive forecasts
  - The predictability of Combination Forecast model suffer greatly from the beginning periods, especially for non-market factors



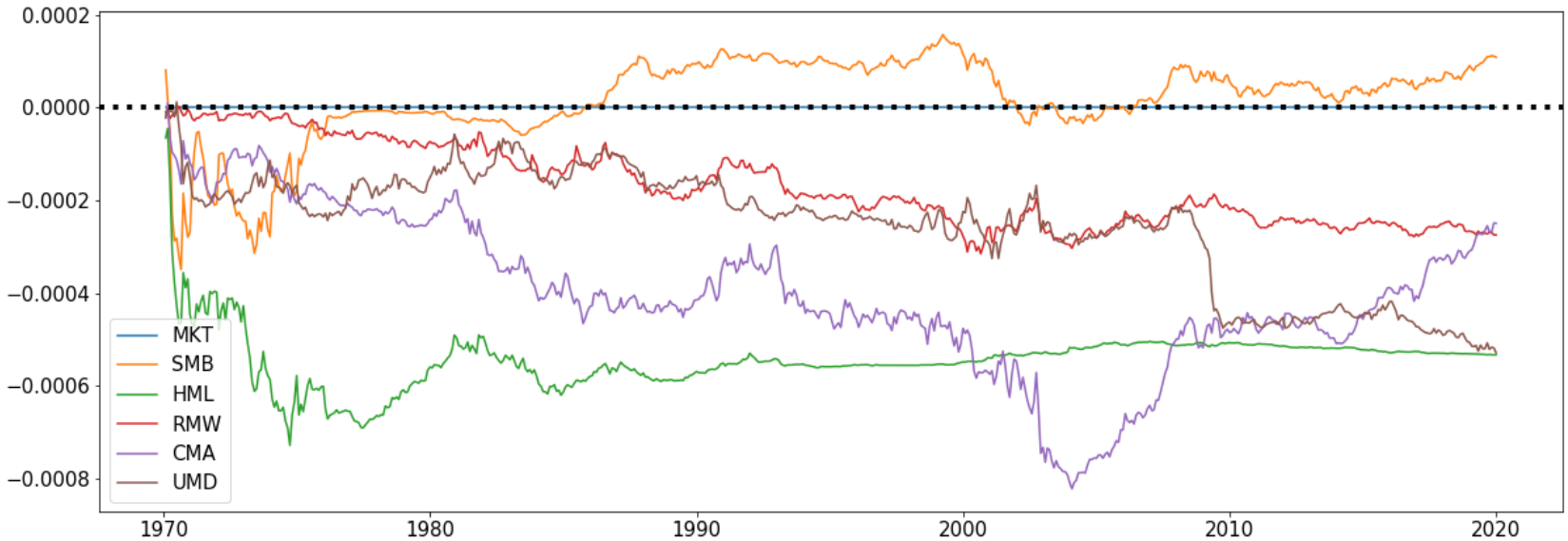
Predictability Visualization

- Difference of cumulative SSE between the Ridge Forecast and the Naive Forecast
  - As another linear model, the predictability also suffers greatly from the beginning periods, but slightly better than the Combination Forecast



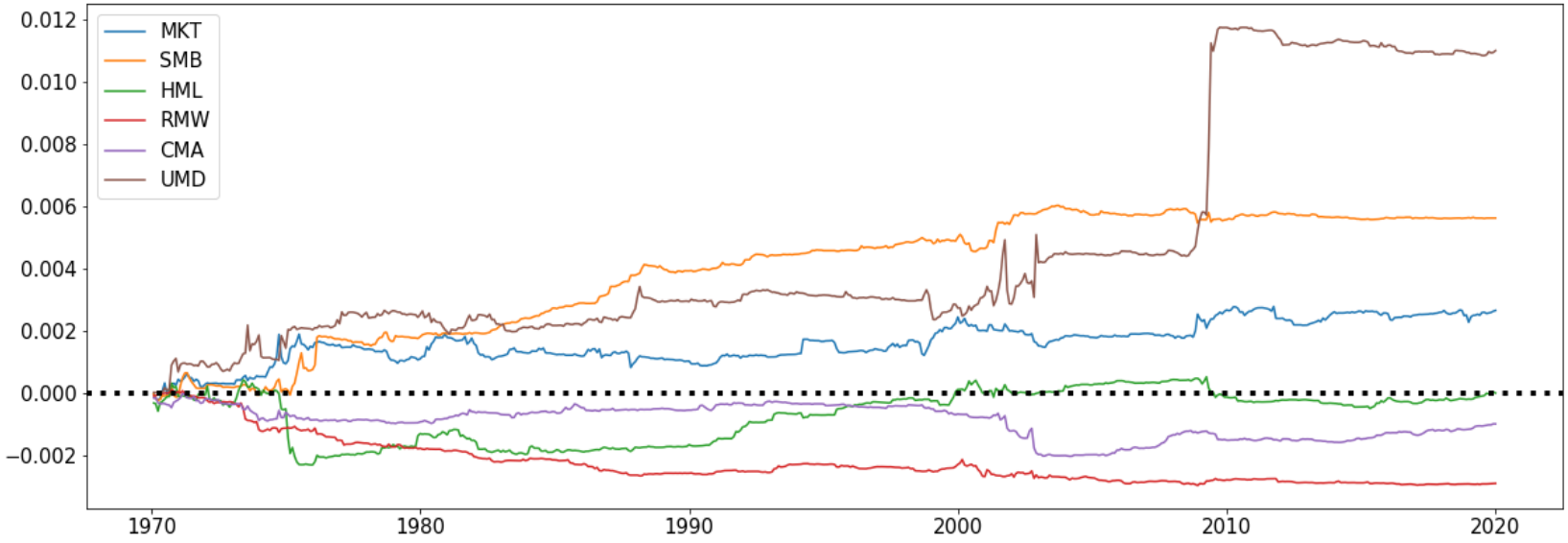
Predictability Visualization

- Difference of cumulative SSE between the Lasso Forecast and the Naive Forecast
  - Better than Combination Forecast and Ridge, however, there are almost no predictability
  - Linear shrinkage predictions are very close to naive forecasts
  - Poor performance of Lasso implies that VAR implied in-sample return predictability is highly overestimated (Lasso is like a standard linear model with well selected variables)



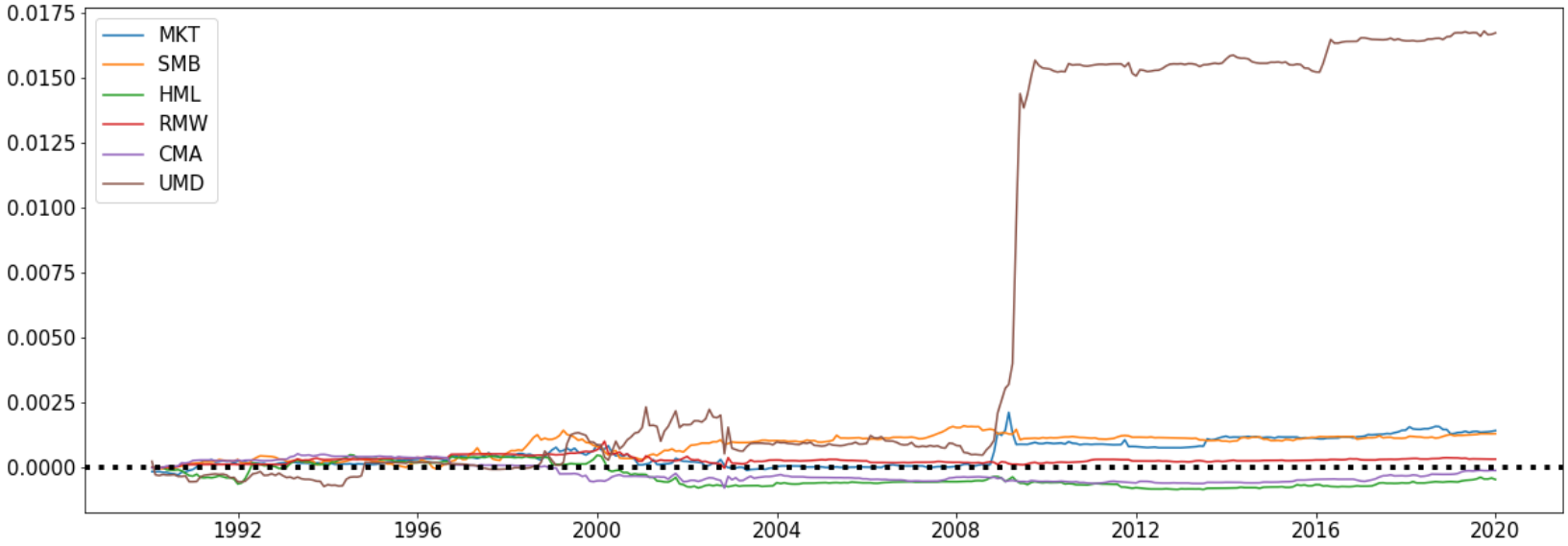
Predictability Visualization

- Difference of cumulative SSE between the BRT Forecast and the Naive Forecast
  - Over the whole sample, BRT obviously show some predictive power for factors MKT, SMB, HML, and UMD
  - Jumps in UMD
  - Nonlinear models outperform linear models
  - Some factors show better predictability than others



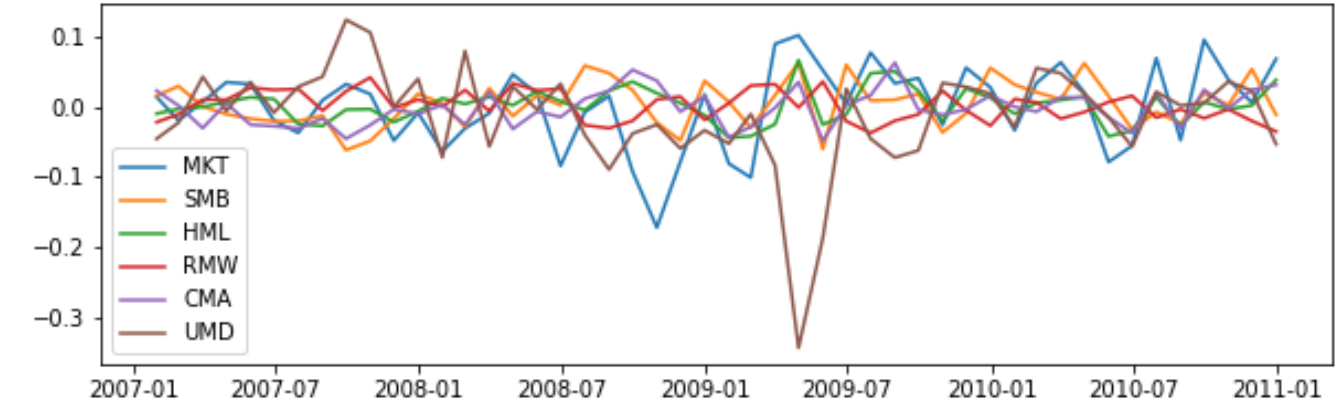
Predictability Visualization

- Difference of cumulative SSE between the Combination Forecast and the Naive Forecast: 1990-2019
  - With nearly half data as training sample, predictive power of Combination Forecast is greatly improved
  - Combination Forecast actually performs the best with over 40 years training sample



Jumps of Momentum Predictability

- Actual factor returns (up figure) and predicted factor returns (down figure)
  - " Momentum crashes are partly forecastable " - Daniel and Moskowitz (2016)



Pseudo-OOS R-squared test: 1970-2019

- Numbers are improved OOS R-squared (%) of one prediction model over another benchmark model, p-value appears in parentheses
  - Overall, BRT is the best model showing predictive power for MKT, SMB, HML, UMD, while it performs best for SMB and UMD
  - The improvements of BRT over linear models could be due to restrictions of small training sample

	MKT	SMB	HML	RMW	CMA	UMD
BRT-Naive	0.19	0.95***	0.11	−0.93***	-0.21	0.69*
	(0.30)	(0.00)	(0.72)	(0.00)	(0.55)	(0.07)
BRT-Comb	0.12	3.17	12.24	13.03	15.07	1.37
	(0.82)	(0.23)	(0.22)	(0.34)	(0.30)	(0.27)
BRT-Ridge	0.11	1.22**	3.24	1.80	3.58	0.58*
	(0.54)	(0.05)	(0.21)	(0.50)	(0.32)	(0.07)
BRT-Lasso	0.11	1.22***	3.24	1.81***	3.58	0.58*
	(0.30)	(0.00)	(0.68)	(0.01)	(0.41)	(0.06)

Pseudo-OOS R-squared test: 2012-2019

- Over 40 years training data, the Combination Forecast has the best performance for all factors
  - Linear relations between equity risk premium and state variables should not be abandoned

	MKT	SMB	HML	RMW	CMA	UMD
BRT-Comb	-0.20	-0.53	-0.23	-0.46	0.17	-1.18
	(0.58)	(0.26)	(0.49)	(0.34)	(0.72)	(0.15)
Comb-Naive	0.67*	0.28	0.56*	0.29	1.66**	1.00
	(0.10)	(0.57)	(0.08)	(0.56)	(0.02)	(0.17)

Basic VAR Results

	Dependent Variables	
Regressors	MKT_t	$E_t[MKT_{t+1}]$
MKT_{t-1}	0.066	-0.004^{***}
$E_{t-1}[MKT_t]$	0.781^{*}	0.974^{***}
R-squared (%)	1.15	93.32
	SMB_t	$E_t[SMB_{t+1}]$
SMB_{t-1}	0.068	-0.07^{***}
$E_{t-1}[SMB_t]$	0.845^{***}	0.986^{***}
R-squared (%)	2.75	95.39
	HML_t	$E_t[HML_{t+1}]$
HML_{t-1}	0.040	-0.004^{***}
$E_{t-1}[HML_t]$	0.068	0.943^{***}
R-squared (%)	0.17	88.76
	RMW_t	$E_t[RMW_{t+1}]$
RMW_{t-1}	0.022	-0.005^{***}
$E_{t-1}[RMW_t]$	-0.060	0.943^{***}
R-squared (%)	0.05	88.63
	CMA_t	$E_t[CMA_{t+1}]$
CMA_{t-1}	0.012	-0.003^{*}
$E_{t-1}[CMA_t]$	0.557^{*}	0.965^{***}
R-squared (%)	0.60	92.52
	UMD_t	$E_t[UMD_{t+1}]$
UMD_{t-1}	0.100^{**}	-0.003^{***}
$E_{t-1}[UMD_t]$	0.454^{**}	0.991^{***}
R-squared (%)	1.89	98.34
Obs	598	

- The predictor chosen in VAR is by BRT which shows the best predictive power over the whole period
- The predictor is highly persistent while the in-sample predictability is weak, showing its potential contributions to DR

Factor Return Variance Decomposition

- Cash flow news drive all six factors
- Correlations between DR and CF is highly positive

	Variance(CF) %	Variance(DR) %	-2Cov(DR, CF) %	Corr(DR, CF)
MKT	180.0	24.7	-104.8	0.785
SMB	276.4	99.3	-275.7	0.832
HML	111.0	0.4	-11.4	0.871
RMW	101.9	0.1	-2.0	0.304
CMA	151.1	18.3	-69.4	0.660
UMD	191.3	32.5	-123.8	0.785

Interpretations for Variance Decomposition

- My results do not support theories suggesting that time-varying risk aversions and investor sentiments cause common DR shocks, which cause most of time-variations in returns, for example, theories in Campbell and Cochrane (1999), Baker and Wurgler (2006)
- Instead, CF shocks, for example, they could be introduced by technology revolutions, are the dominant cause of factor return movements
- Positive correlations between CF and DR is consistent with Lettau and Ludvigson (2005), which explains the positive correlations as that expected dividend growth goes up as risk premia go higher during recessions

Correlations of Market DR and CF with Non-market DR and CF

- Generally, low correlations suggest that shocks driving market factor and non-market factors are distinct
- Magnitude of correlations between MKT CF and Non-MKT CF are slightly higher than those correlations with DR, suggesting that different factor returns are not driving by common discount rate shocks

	MKT CF		MKT DR	
	Non-MKT CF	Non-MKT DR	Non-MKT CF	Non-MKT DR
SMB	0.282	0.332	0.148	0.171
HML	-0.271	-0.171	-0.172	-0.130
RMW	-0.114	0.056	-0.117	0.006
CMA	-0.212	-0.081	-0.137	-0.132
UMD	-0.126	-0.056	-0.058	0.006

Correlations of CF and DR with Macroeconomic Shocks

- Correlations among factor returns' DR, CF and macroeconomic shocks are also examined
  - Macroeconomics shocks are obtained as shocks of AR(1) process for consumption growth, employment growth, term spread, and default spread
- All their correlations are very low, while market factor components show slightly higher correlations with consumption and employment shocks in magnitude
- Market factors, non-market factors and macroeconomics experience distinct shocks

Correlations of CF and DR with Macroeconomic Shocks

		Consumption	Employment	Term Spread	Default Spread
CF	MKT	-0.101	0.074	0.067	-0.006
	SMB	-0.073	-0.002	0.038	0.037
	HML	-0.018	0.049	0.065	-0.012
	RMW	0.004	-0.030	-0.078	-0.011
	CMA	-0.033	-0.010	0.059	0.076
	UMD	0.036	0.033	-0.051	-0.112
DR	MKT	-0.089	0.121	0.040	0.056
	SMB	-0.087	0.025	0.013	0.142
	HML	-0.021	0.058	0.110	0.023
	RMW	-0.084	0.004	-0.005	0.020
	CMA	-0.072	0.009	0.083	0.063
	UMD	0.011	0.018	-0.040	-0.137

Conclusion

- Using machine learning predictive models, I examine factor predictability. While nonlinear models predict returns better, evidence suggests that linear models should not be abandoned
- I examine the time-varying sources of asset pricing factors by a bivariate VAR with an nontrivial predictor
- I find two facts counter to previous empirical results which are from VAR with standard predictors
  - Cash flow news drive all factor returns
  - Cash flow news correlate positively with discount rate news
- Cash flow news and discount rate news driving market factor, non-market factors, and macroeconomics are different